IT Ticketing & SLA Simulation Project Report (Jira)

As part of a self-led learning initiative, I completed a hands-on IT ticketing and SLA (Service-Level Agreement) simulation project using **Jira**. The goal was to mimic real-world IT support operations by creating, resolving, and tracking technical issues, while applying custom SLA resolution targets and learning how ticket workflows are managed in professional environments.

Since I used the **free version of Jira**, I worked around its limitations by manually simulating SLA rules. Instead of using built-in SLA timers (which require a paid plan), I added a **custom field for due dates** based on ticket priority, and applied **custom labels** (`sla-met`, `sla-breached`) to indicate whether a ticket was resolved within its SLA timeframe. This helped reinforce the concept of time-based resolution goals and how they impact IT support performance tracking.

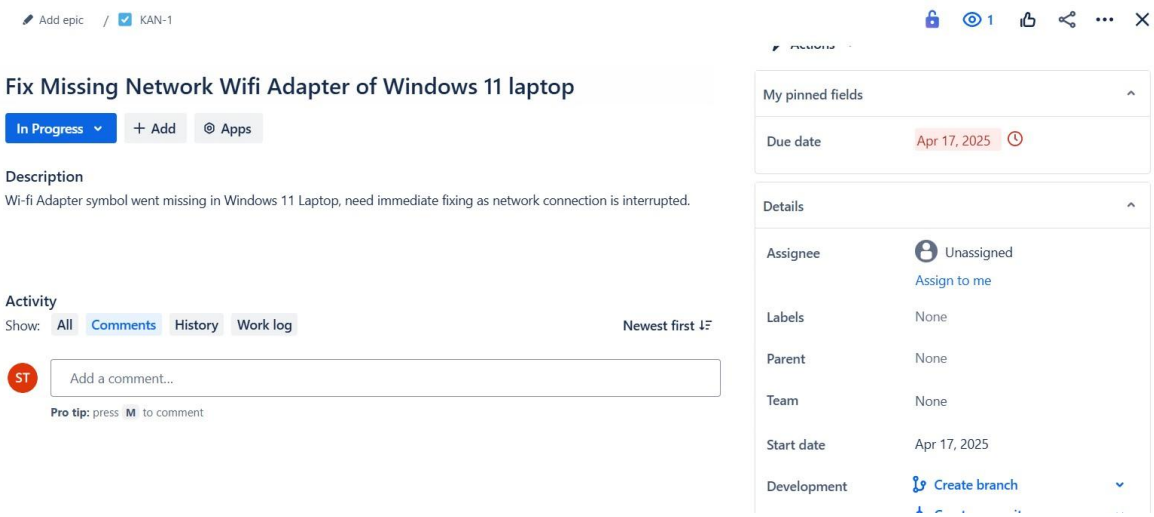

Image 1 : Generating ticket

During the project, I created and resolved three realistic IT support tickets:

1. The first ticket involved a **Windows 11 laptop not detecting its Wi-Fi adapter**. I documented the issue, added relevant details, and resolved it by scanning for hardware changes. The issue was fixed successfully the same day, within the assigned SLA window. I labeled the ticket as `sla-met` to reflect on-time resolution.

2. The second ticket focused on a **VMware networking issue**, where **Kali Linux running inside VMware could not communicate with a penetration testing**

target. I categorized it as a medium-priority network issue with a 1-day resolution deadline. After troubleshooting the VM's network adapter settings, I restored communication, documented the fix, and labeled the ticket appropriately as `sla-met`, `network-issue`, and `medium-priority`.

3. In the third scenario, I simulated a **"Laptop Not Powering On" issue**. The laptop was unresponsive and would not boot. I performed basic hardware-level troubleshooting steps like unplugging all peripherals, holding the power button to discharge static, and verifying power supply connections. The issue was resolved through a hard reset within 4 hours, meeting the high-priority SLA target. I marked this ticket with labels such as `sla-met`, `hardware-issue`, and `high-priority`.

To enhance tracking and reporting, I created several **custom JQL filters** in Jira. These allowed me to query and view tickets based on SLA status, creation date, resolution state, and priority. For example, I saved filters for:

All tickets that met or breached SLA deadlines (`SLA Tickets by Priority`)

Tickets that missed SLA targets (`SLA Breached Tickets`)

Resolved issues (`Resolved IT Tickets`)

Tickets created within the last 7 days (`Tickets Created This Week`)

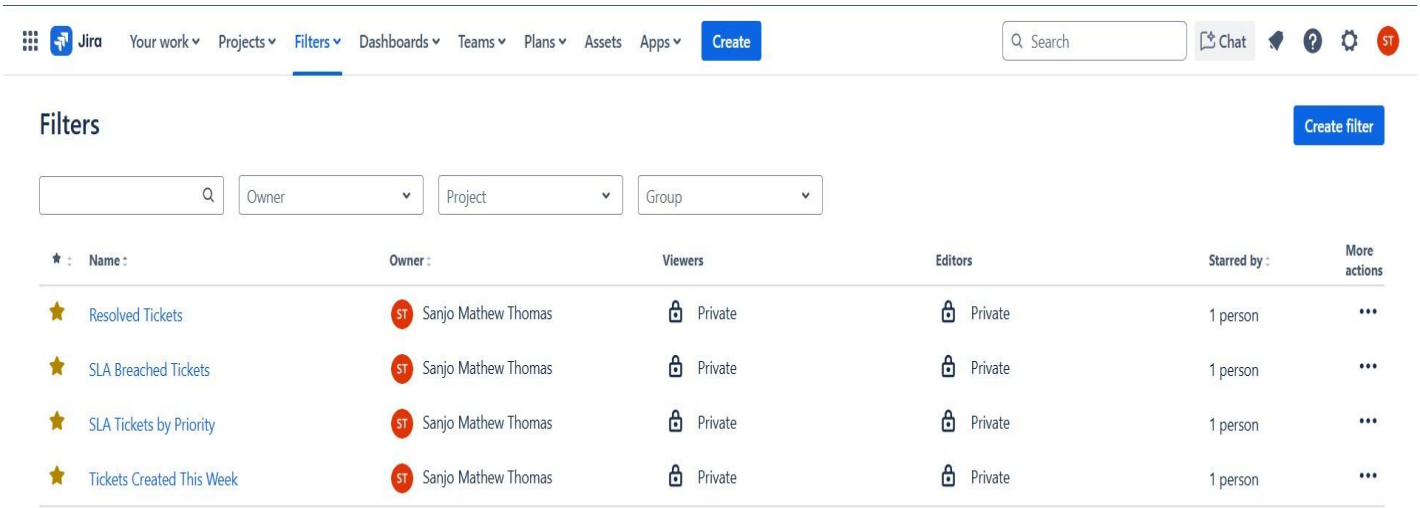

Image 2: Filters List

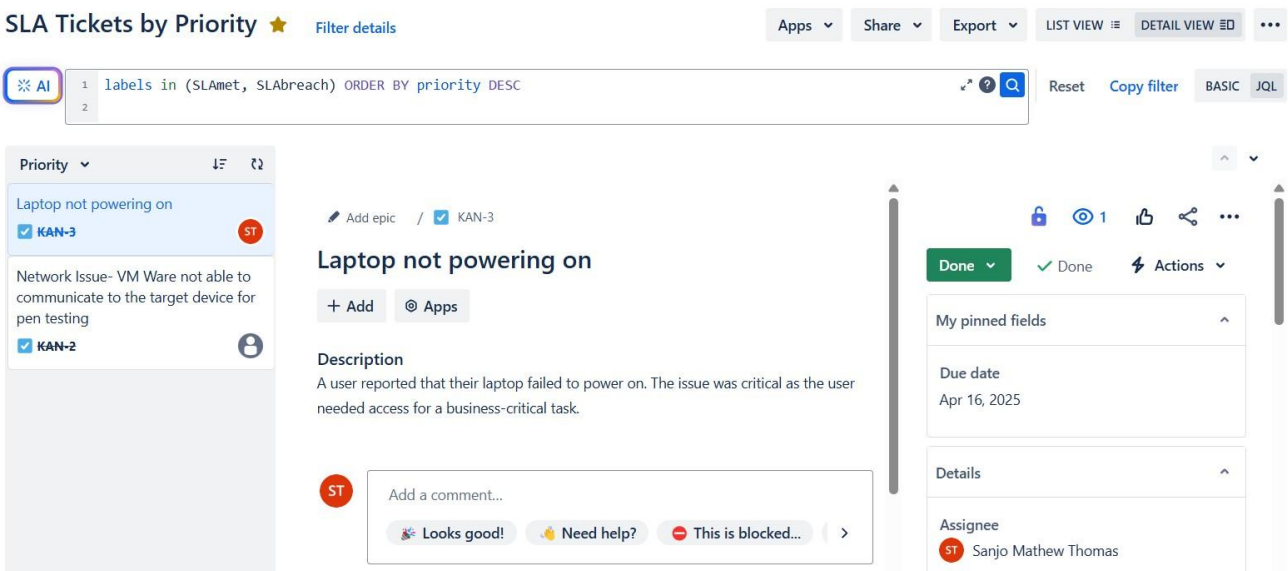

Image 3: JQL Query for tickets by priority

Although I couldn't access Jira's dashboard feature due to UI restrictions and billing requirements, I replicated dashboard functionality by using saved filters, viewing result lists, and capturing screenshots of ticket states. This gave me a practical understanding of how service desk agents and IT support teams monitor performance metrics and issue resolution rates.

This project gave me hands-on exposure to core IT support and ticketing concepts, including **ticket creation, prioritization, resolution documentation, SLA simulation, and basic JQL filtering**. It helped me understand how service desks operate in a structured ticketing environment and how SLAs are tracked for accountability. Despite being new to Jira and ticketing systems in general, I was able to build a lightweight yet realistic simulation from scratch.